

JK 12-16 Python First Programmer
Lesson 04 : Conditional Statements

Duration : 90 mins

Description

Students learn what are conditional statements and how to use if, elif and else functions. They create a few simple projects and two bigger projects, including a “username checker” which checks if a username given by a user is following the rules.

Objectives

Students will be able to:

1. Understand what is conditional statements and apply it in a code.
2. Explain the difference between if, elif, and else and use it in a code.
3. Practice how to combine conditional statements with logical operators in a code.
4. Analyze code that incorporate conditional statements and the outcome of the code.
5. Create projects combining all concepts learned during the lesson.

Tools

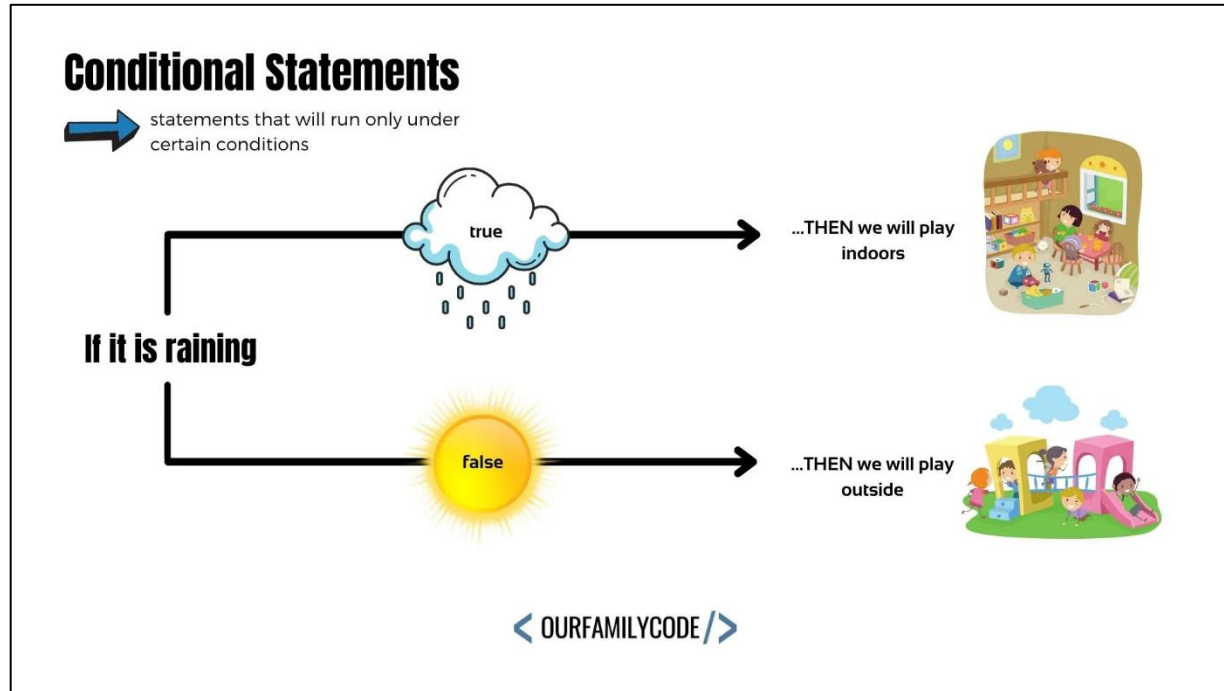
1. Pycharm

Class Structure

Duration	Topic
5 mins	Preparation
5 mins	Comprehension Check
15 mins	Conditional Statements - If
10 mins	If-Else
10 mins	If-Elif-Else
15 mins	Traffic Light
15 mins	Conditional Statements with Logical Operators
10 mins	Review Quiz
5 mins	Wrap Up

Planned Activities	Resources
Preparation (5 mins)	
- Teacher greets students and ask how's their conditions. - Instruct students to open their pycharm. - If possible, teacher can incorporate a mini game during this preparation time to engage with the students.	
Comprehension Check (5 mins)	
- Teacher ask students if they are still remember about the materials from the previous lesson. - Teacher check on students' comprehension regarding the previous lesson by asking relatable questions. - Teacher explain and demonstrate in brief or summarize about what they have learned from the previous lesson.	
Conditional Statements - If (15 mins)	
Explain - We explain to students that a conditional statement is a programming language feature to decide whether it's true or not, then do the action later on. - If a certain conditional is fulfilled/true, the action within that conditional statement will be executed by the program.	

- We can give an example: the conditional statement is about weather. If it's raining, then we need to bring an umbrella.



- We explain to students that for the conditional statements, we'll use it with comparison operators that we've learned in the previous lesson. We can use the equal, greater than, less than, not equal, and many more. Whenever the condition is true, the statements will be executed by the program.
- We show students how to write the conditional statements. Explain to students that we need to put the condition after the if, and don't forget to put the colon (:) mark after the condition.
- After the colon mark, enter and press tab on the keyboard so it will be one column to the right. Thus, the python program will know that the statements/action they need to execute is the row on the right side after the condition row.

Explain and Code – If Project

- We show the students the project. We tell the students that in this project it has a variable, we put a random number from 1 to 100. If the number is less than 50, the program will print a message that says “Yes! The random number is less than 50”.
- We tell students to create a new python file and create a variable and put a random number from 1-100 to it by themselves. Also, remind students because we’re using random so we need to import random from the library at the beginning.
- Remind students to display the intro or instruction, so the user knows what to do with the program.
- Teachers may give hints to students if they don’t remember how to declare variables or make a random number.

File/Code:

[1.lf.py](#)

Explain and Code – If Conditional

- Create the conditional statement together with students.
- After finishing the first conditional project, ask students to change the comparison operators. They may change it into equal, not equal, greater than, and so on. Remind students to also change the statement message if they change the comparison operators.
- Ask the student what happened with the result.

If-Else (10 mins)

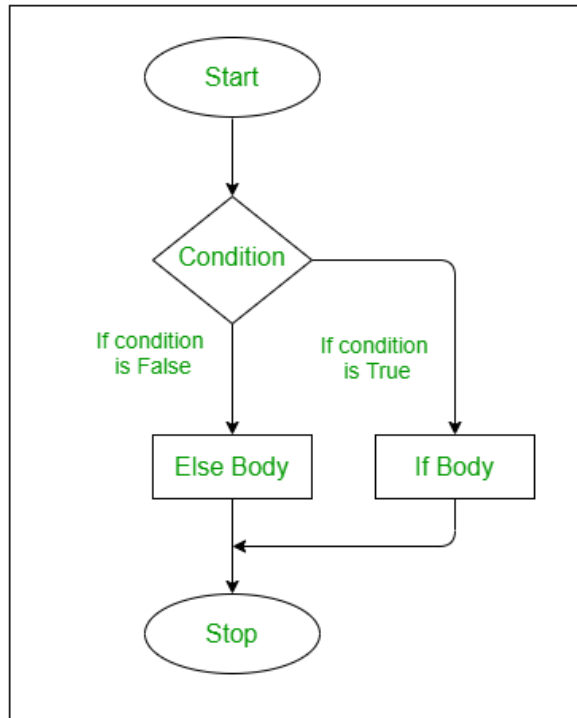
Explain and Code – If-Else

- We tell students that we can create statements when the condition is not fulfilled or not true. We can add it with ELSE.
- Explain to students that ELSE block will be run when the if block condition is not satisfied or evaluated to be False.

File/Code:

[2.If-Else.py](#)

- We tell students that we are going to upgrade their project so when the number is not less than 50, we will display a message.
- Show students how to add the ELSE.



If-Elif-Else (10 mins)

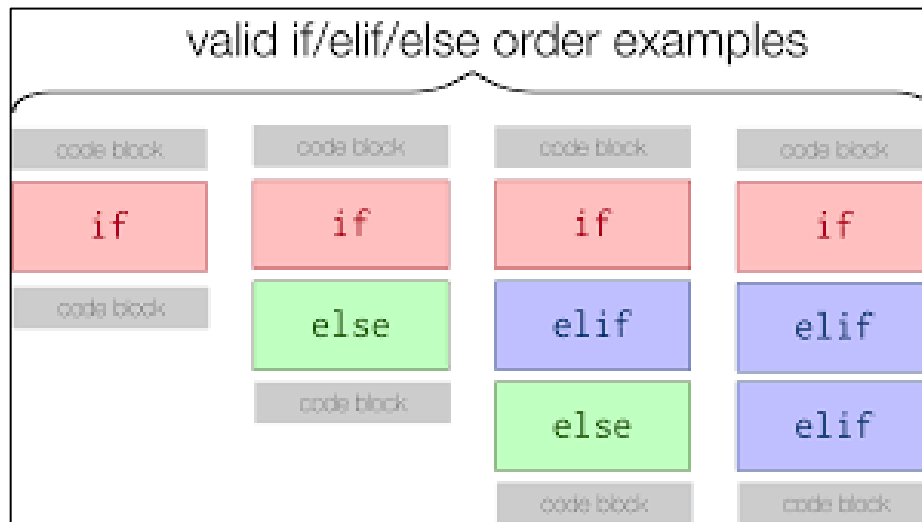
Explain and Code – If-Elif-Else

- We ask students, what if we have more than one condition? Explain to students that we can put an ELIF statement if we have more than one condition.
- We show students how to add one more condition with the ELIF statement.

File/Code:

[3.If-Elif-Else.py](#)

- Show students that ELIF can be as much as needed. It means that the conditions that is going to be checked can be more than one.
- Remind students that ELIF cannot stand alone without IF. We'll always start the conditional statement with an IF statement. We can add more later with ELIF or end it with ELSE.



Traffic Light (15 mins)

Explain and Show – Traffic Light

- We show the traffic light project students. We explain to students that in this project, we need to put the color of the traffic light.
- If we put a different color, it will also display a different message.
- Explain that to compare the string, we need to put an equal sign (`==`) to compare the input and the color word.
- Let students do it by themselves and the teacher may give them hints if they don't understand some parts.

File/Code:

[4.TrafficLight.py](#)

Conditional Statements with Logical Operators (15 mins)

Explain – Logical Operator

- We explain to students that conditional statements and logical operators can be combined.
- In several cases, we need to combine it because we need to check 2 or more conditions at the same time.

Explain and Show – Logical Operator (AND)

- We show students the 5.And.py project.
- We explain to students that the true statement will be executed only if both of the conditions are true.
- Change the quantity of the fruit or change the fruit name and show the result to the students.

File/Code:

[5.And.py](#)

Explain and Show – Logical Operator (OR)

- We show students the Or.py project.
- We explain to students that the true statement will be executed if one of the conditions is true.
- If both of the conditions are not true, the false statement will be displayed.
- Change the fruit name to be something else and show the result to students.

File/Code:

[6.Or.py](#)

Explain and Show – Username Checker

- We show the projects to students. We explain to students that it's a program to check if the username is already up to the program's standards or not.
- User needs to put at least 6 characters but no more than 10 characters. Also, the user needs to put the username in all lowercase.
- We can tell students that to know the length of the string, we use the len(string) function.

File/Code:

[7.UsernameChecker.py](#)

- In addition, we can add other conditions. We explain, the condition is that the username cannot use lowercase letters. This can be achieved using the `.islower()` method. This condition will be placed inside the elif block.
- Explain to students, the elif block is a decision that has more than one condition and can be used to avoid nested if.
- Teachers may give hints to students that we need logical operators to check all of the conditions at the same time.

Review Quiz (10 mins)

Individual Activity – Review Quiz

- We show students the review quiz questions for today and give them some time to answer it by themselves in a notepad file.
- When everyone is done, we tell them to save their answers and review it together.

Wrap Up (5 mins)

Wrapping Up

- We sum up by asking questions, making sure the students remember the most important things.
- We thank students for doing a great job overall.

☐ ☆ Good Luck With The Class !! ☆ ☐